

संभव BATCH

Under the Guidance of Vipulendra Tripathi Sir

**Geography: Atmosphere, Clouds, Rainfall, Water Cycle & Trade Winds | SSC CGL
Hard Level MCQs**

Q1. Consider the following statements regarding the Earth's atmosphere:

1. The thickness of the troposphere is maximum at the equator.
2. The stratosphere contains the ozone layer which protects the Earth from harmful ultraviolet radiation.
3. Radio waves transmitted from the Earth are reflected back by the mesosphere.

Which of the statements given above is/are correct?

[SSC CGL Tier-II 2023]

(A) 1, 2 and 3

(B) 1 and 3 only

(C) 1 and 2 only

(D) 2 and 3 only

Q2. Assertion (A): The temperature in the stratosphere increases with an increase in altitude.

Reason (R): The absorption of ultraviolet radiation by the ozone layer is responsible for this temperature inversion.

[SSC CGL Tier-I 2022]

(A) Both A and R are true and R is the correct explanation of A.

(B) A is false but R is true.

(C) A is true but R is false.

(D) Both A and R are true but R is NOT the correct explanation of A.

Q3. Which of the following layers of the atmosphere is virtually free from clouds and associated weather phenomena, making it ideal for flying jet aircraft?

[SSC CHSL Tier-I 2021]

(A) Stratosphere

(B) Troposphere

(C) Thermosphere

(D) Mesosphere

Q4. Match List I (Atmospheric Gas) with List II (Percentage by Volume):

List I	List II
A. Nitrogen	1. 0.93%
B. Oxygen	2. 0.03% (historically, pre-industrial)
C. Argon	3. 20.95%
D. Carbon Dioxide	4. 78.08%

(A) A-3, B-4, C-1, D-2

(B) A-4, B-3, C-2, D-1

(C) A-4, B-3, C-1, D-2

(D) A-4, B-1, C-3, D-2

Q5. Consider the following statements regarding the Ionosphere:

1. It is located between 80 km and 400 km above the Earth's surface.
2. The Aurora Borealis and Aurora Australis phenomena occur in this layer.
3. Temperature starts decreasing rapidly in this layer with height.

Which of the statements given above is/are NOT correct?

[SSC CGL 2021]

(A) 2 only

(B) 1 only

(C) 3 only

(D) 1 and 3 only

Q6. Assertion (A): The amount of insolation received at the Earth's surface decreases from the equator towards the poles.

Reason (R): The angle of incidence of the sun's rays becomes more oblique towards the poles.

[SSC MTS 2022]

(A) Both A and R are true and R is the correct explanation of A.

(B) A is false but R is true.

(C) A is true but R is false.

(D) Both A and R are true but R is NOT the correct explanation of A.

Q7. Which of the following statements about the 'Normal Lapse Rate' is correct?

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- (A) Temperature decreases at a rate of 1°C for every 1000 meters of ascent in the stratosphere.
- (B) Temperature increases uniformly throughout the mesosphere.
- (C) Temperature increases at a rate of 1°C for every 165 meters of ascent in the troposphere.
- (D) Temperature decreases at a rate of 1°C for every 165 meters of ascent in the troposphere.

Q8. Consider the following components of the atmosphere:

1. Water vapour
2. Dust particles
3. Ozone

Which of the above are highly variable in the lower atmosphere (troposphere)?

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- (A) 2 and 3 only
- (B) 1 and 3 only
- (C) 1, 2 and 3
- (D) 1 and 2 only

Q9. Match List I (Atmospheric Layer) with List II (Characteristic Feature):

List I	List II
A. Troposphere	1. Meteorites burn up here
B. Stratosphere	2. Electrically charged particles
C. Mesosphere	3. Contains most of the atmospheric mass
D. Thermosphere	4. Mother of pearl clouds (Nacreous)

(A) A-3, B-4, C-2, D-1**(B)** A-3, B-4, C-1, D-2**(C)** A-4, B-3, C-1, D-2**(D)** A-1, B-2, C-3, D-4**Q10. Assertion (A): Carbon dioxide is meteorologically a very important gas.****Reason (R): It is transparent to the incoming solar radiation but opaque to the outgoing terrestrial radiation.****[SSC CHSL Tier-I 2022]****(A)** A is false but R is true.**(B)** Both A and R are true but R is NOT the correct explanation of A.**(C)** Both A and R are true and R is the correct explanation of A.**(D)** A is true but R is false.

Q11. Which of the following statements regarding Albedo of the Earth is NOT correct?

[SSC CGL Tier-II 2021]

- (A) Deserts have a higher albedo than dense forests.
- (B) Fresh snow has a lower albedo compared to a thick cloud cover.
- (C) It is the proportion of shortwave solar radiation reflected back into space by the Earth's surface and atmosphere.
- (D) The average albedo of the Earth is approximately 30-35%.

Q12. Consider the following statements about the Heat Budget of the Earth:

1. Out of 100 units of incoming solar radiation, exactly 50 units reach the Earth's surface.
2. The Earth radiates energy back into space in the form of long-wave terrestrial radiation.
3. The atmosphere is primarily heated directly by incoming short-wave solar radiation.

Which of the statements given above is/are correct?

[SSC CGL 2021]

- (A) 2 and 3 only
- (B) 1, 2 and 3
- (C) 1 and 2 only
- (D) 2 only

Q13. Assertion (A): Cirrus clouds are always composed of ice crystals.

Reason (R): They are formed at very high altitudes (8,000 - 12,000 meters) where temperatures are completely sub-zero.

[SSC CPO 2022]

(A) Both A and R are true but R is NOT the correct explanation of A.

(B) Both A and R are true and R is the correct explanation of A.

(C) A is false but R is true.

(D) A is true but R is false.

Q14. Match List I (Cloud Type) with List II (Characteristics):

List I	List II
A. Cirrus	1. Cauli-flower shaped, vertical development
B. Stratus	2. Dark, grey, rain-bearing continuous layer
C. Cumulus	3. Detached, feathery appearance, high altitude
D. Nimbostratus	4. Layered, covering large portions of the sky

(A) A-3, B-1, C-4, D-2

(B) A-4, B-3, C-2, D-1

(C) A-2, B-4, C-1, D-3

(D) A-3, B-4, C-1, D-2

Q15. Which of the following cloud types is primarily responsible for severe thunderstorms, hail, and tornadoes?

[SSC CHSL 2023]

(A) Cirrocumulus

(B) Altostratus

(C) Cumulonimbus

(D) Stratocumulus

Q16. Consider the following statements:

1. Altocumulus and Altostratus are classified as middle-altitude clouds.
2. Stratus clouds are formed by the loss of heat or the mixing of air masses with different temperatures.
3. Nimbostratus clouds are extremely high-altitude clouds associated with fair weather.

Which of the statements given above is/are NOT correct?

[SSC CGL Tier-II 2022]

(A) 2 and 3 only

(B) 3 only

(C) 1, 2 and 3

(D) 1 and 2 only

Q17. Assertion (A): A halo ring around the sun or the moon is typically associated with Cirrostratus clouds.

Reason (R): Cirrostratus clouds consist of ice crystals which refract the incoming light to form halos.

[SSC MTS 2023]

- (A) A is false but R is true.
- (B) Both A and R are true and R is the correct explanation of A.
- (C) Both A and R are true but R is NOT the correct explanation of A.
- (D) A is true but R is false.

Q18. Which of the following cloud types produces a 'mackerel sky', resembling the scales of a fish?

[SSC CGL Tier-I 2021]

- (A) Cirrocumulus
- (B) Cumulus
- (C) Altostratus
- (D) Nimbostratus

Q19. Consider the following pairs (Cloud Prefix vs Meaning):

1. Cirro - High altitude
2. Alto - Middle altitude
3. Nimbo - Rain bearing
4. Strato - Vertical growth

How many of the above pairs are correctly matched?

[SSC CPO 2021]

(A) All four pairs

(B) Only one pair

(C) Only three pairs

(D) Only two pairs

Q20. Which among the following clouds is lowest in the atmosphere?

[SSC CHSL Tier-I 2022]

(A) Cirrostratus

(B) Cirrus

(C) Altocumulus

(D) Stratus

Q21. Assertion (A): Fog is essentially a stratus cloud that forms near the ground.

Reason (R): It forms when the air near the ground cools enough to turn its water vapour into liquid water or ice.

[SSC CGL 2020]

(A) A is false but R is true.

(B) A is true but R is false.

(C) Both A and R are true and R is the correct explanation of A.

(D) Both A and R are true but R is NOT the correct explanation of A.

Q22. Which of the following statements about noctilucent clouds is correct?

[SSC CGL Tier-II 2023]

- (A) They are entirely composed of volcanic ash.
- (B) They form in the mesosphere and are visible in a deep twilight.
- (C) They are the primary cause of acid rain in tropical regions.
- (D) They are the lowest clouds in the troposphere.

Q23. Consider the following statements regarding types of rainfall:

1. Convictional rainfall is common in the equatorial regions.
2. Orographic rainfall occurs on the leeward side of the mountains.
3. Cyclonic rainfall is common in the mid-latitudes due to the meeting of warm and cold air masses.

Which of the statements given above is/are correct?

[SSC CGL Tier-I 2023]

- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1, 2 and 3
- (D) 1 and 3 only

Q24. Assertion (A): The windward side of the Western Ghats receives heavy rainfall while the Deccan Plateau receives much less.

Reason (R): The Deccan Plateau lies in the rain-shadow region of the Western Ghats.

[SSC CGL 2023]

- (A) A is true but R is false.
- (B) Both A and R are true and R is the correct explanation of A.
- (C) Both A and R are true but R is NOT the correct explanation of A.
- (D) A is false but R is true.

Q25. Match List I (Rainfall Feature) with List II (Type of Rainfall):

List I	List II
A. Frontal activity	1. Convectional Rainfall
B. Rain shadow area	2. Orographic Rainfall
C. Torrential downpour with lightning	3. Cyclonic/Frontal Rainfall

(A) A-3, B-2, C-1

(B) A-2, B-3, C-1

(C) A-1, B-2, C-3

(D) A-3, B-1, C-2

Q26. Which of the following statements about 'Acid Rain' is NOT correct? [SSC CHSL 2021]

(A) It is exclusively a phenomenon of the upper stratosphere.

(B) It occurs when these gases react with water, oxygen, and other chemicals in the atmosphere.

(C) It can lower the pH of lakes and streams, harming aquatic life.

(D) It is primarily caused by emissions of sulfur dioxide and nitrogen oxides.

Q27. Consider the following features:

- 1. Rapid heating of the earth's surface.**
- 2. Expansive upward movement of warm air.**
- 3. Heavy but short-duration showers in the afternoon.**

The above features describe which type of rainfall?

[SSC MTS 2022]

(A) Frontal rainfall

(B) Orographic rainfall

(C) Monsoonal rainfall

(D) Convectional rainfall

Q28. Assertion (A): Rainfall generally decreases from the equator towards the poles.

Reason (R): The temperature and moisture-holding capacity of the air continuously decreases from the equator to the poles.

[SSC CGL Tier-II 2021]

(A) Both A and R are true but R is NOT the correct explanation of A.

(B) Both A and R are true and R is the correct explanation of A.

(C) A is false but R is true.

(D) A is true but R is false.

Q29. Which of the following regions is NOT known for receiving maximum convectional rainfall?

[SSC CGL 2020]

(A) The Congo Basin

(B) The Mediterranean region

(C) The South-East Asian islands

(D) The Amazon Basin

Q30. Regarding the distribution of rainfall globally, match the regions:

List I (Region)	List II (Rainfall Regime)
A. Equatorial region	1. Winter rainfall
B. Mediterranean region	2. Heavy rainfall throughout the year
C. Monsoon region	3. Summer rainfall
D. Deserts	4. Scanty rainfall

(A) A-1, B-2, C-3, D-4

(B) A-2, B-1, C-3, D-4

(C) A-3, B-1, C-2, D-4

(D) A-2, B-3, C-1, D-4

Q31. Assertion (A): The amount of rainfall generally decreases as one goes from the coast to the interior of continents.

Reason (R): The moisture-bearing winds lose most of their moisture over the coastal areas before reaching the interior.

[SSC CHSL Tier-I 2022]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q32. Which of the following is NOT a necessary condition for precipitation to occur?

[SSC CGL Tier-I 2023]

(A) Presence of condensation nuclei

(B) Saturation of the air mass

(C) Cooling of the air mass

(D) Presence of an anticyclone

Q33. Consider the following statements regarding Humidity:

1. Absolute humidity is expressed in grams per cubic meter of air.
2. Relative humidity is highest when the air temperature is highest.
3. Specific humidity remains constant even if the volume of air expands due to pressure changes.

Which of the statements given above is/are correct?

[SSC CGL Tier-II 2023]

(A) 1 and 3 only

(B) 1 and 2 only

(C) 2 and 3 only

(D) 1, 2 and 3

Q34. Assertion (A): Evaporation is a cooling process.

Reason (R): It requires latent heat of vaporization, which is drawn from the surroundings, thereby lowering the surrounding temperature.

[SSC CPO 2021]

(A) A is true but R is false.

(B) A is false but R is true.

(C) Both A and R are true and R is the correct explanation of A.

(D) Both A and R are true but R is NOT the correct explanation of A.

Q35. Match List I (Process in Water Cycle) with List II (Definition):

List I	List II
A. Sublimation	1. Water vapor turning directly into ice
B. Transpiration	2. Release of water vapor by plants
C. Condensation	3. Ice turning directly into water vapor
D. Deposition	4. Gas turning into a liquid

(A) A-1, B-2, C-4, D-3

(B) A-3, B-4, C-2, D-1

(C) A-3, B-2, C-4, D-1

(D) A-1, B-4, C-2, D-3

Q36. Which of the following statements about Dew Point is correct?

[SSC CHSL 2023]

(A) It is the temperature to which air must be cooled to become saturated with water vapor.

(B) It is the point in the atmosphere where clouds begin to form.

(C) It is the temperature at which water boils at high altitudes.

(D) It is always below the freezing point of water.

Q37. Assertion (A): During the winter season, relative humidity is generally higher in the early morning compared to the afternoon.

Reason (R): As temperature increases in the afternoon, the moisture-holding capacity of the air increases, reducing relative humidity.

[SSC MTS 2021]

(A) Both A and R are true but R is NOT the correct explanation of A.

(B) A is false but R is true.

(C) Both A and R are true and R is the correct explanation of A.

(D) A is true but R is false.

Q38. Consider the following statements:

1. Evaporation is inversely proportional to the temperature of the water and air.
2. High wind speed increases the rate of evaporation.
3. An increase in the salinity of water decreases the rate of evaporation.

Which of the statements given above are correct?

[SSC CGL Tier-I 2021]

(A) 1 and 2 only

(B) 2 and 3 only

(C) 1 and 3 only

(D) 1, 2 and 3

Q39. The instrument used to measure relative humidity is called:

[SSC CPO 2020]

(A) Barometer

(B) Anemometer

(C) Lysimeter

(D) Hygrometer

Q40. Which of the following factors does NOT significantly affect the rate of transpiration in plants?

[SSC CGL Tier-II 2022]

(A) Light intensity

(B) Wind speed

(C) Soil salinity

(D) Atmospheric pressure

Q41. Consider the following statements regarding the Coriolis Force:

1. It deflects winds to the right in the Northern Hemisphere.
2. It is maximum at the equator and zero at the poles.
3. It is caused by the rotation of the Earth on its axis.

Which of the statements given above is/are NOT correct?

[SSC CGL Tier-II 2023]

(A) 1 and 3 only

(B) 1 only

(C) 2 only

(D) 3 only

Q42. Assertion (A): The Inter-Tropical Convergence Zone (ITCZ) is a low-pressure zone located at the equator.

Reason (R): Trade winds of the Northern and Southern Hemispheres converge at this zone, forcing the air to rise.

[SSC CGL Tier-I 2022]

(A) A is false but R is true.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) Both A and R are true and R is the correct explanation of A.

(D) A is true but R is false.

Q43. Match List I (Planetary Winds) with List II (Latitudinal Zone):

List I	List II
A. Doldrums	1. 30° - 60° N and S
B. Trade Winds	2. 0° - 5° N and S
C. Westerlies	3. 5° - 30° N and S
D. Polar Easterlies	4. 60° - 90° N and S

(A) A-3, B-2, C-4, D-1

(B) A-3, B-2, C-1, D-4

(C) A-2, B-3, C-4, D-1

(D) A-2, B-3, C-1, D-4

Q44. Which of the following statements about the 'Roaring Forties' is correct?

[SSC CGL Tier I 2021]

- (A) They are seasonal monsoon winds affecting the Indian subcontinent.
- (B) They are strong westerly winds found in the Southern Hemisphere between 40° and 50° latitudes.
- (C) They are intense tropical cyclones in the Indian Ocean.
- (D) They are powerful trade winds blowing in the Northern Hemisphere.

Q45. Assertion (A): The sub-tropical high-pressure belts are also known as 'Horse Latitudes'.

Reason (R): They are characterized by calm, light, and variable winds due to the subsidence of air.

[SSC CGL 2020]

- (A) Both A and R are true but R is NOT the correct explanation of A.
- (B) A is false but R is true.
- (C) Both A and R are true and R is the correct explanation of A.
- (D) A is true but R is false.

Q46. Consider the following statements:

1. Trade winds blow from Sub-tropical high-pressure belts to Equatorial low-pressure belts.
2. In the Northern Hemisphere, Trade winds blow from the North-East direction.
3. They are known for being highly erratic and seasonal.

Which of the statements given above is/are correct?

[SSC CGL Tier-II 2021]

(A) 1 and 3 only

(B) 1 and 2 only

(C) 1, 2 and 3

(D) 2 and 3 only

Q47. Which of the following pressure belts is dynamically induced rather than thermally induced?

[SSC MTS 2022]

(A) Polar High-Pressure Belt

(B) Sub-Tropical High-Pressure Belt

(C) Equatorial Low-Pressure Belt

(D) None of the above

Q48. Match List I (Local Wind) with List II (Region):

List I	List II
A. Chinook	1. Alps (Europe)
B. Foehn	2. Rockies (North America)
C. Loo	3. Sahara Desert (Africa)
D. Harmattan	4. Northern Plains (India)

(A) A-1, B-2, C-4, D-3

(B) A-1, B-2, C-3, D-4

(C) A-2, B-1, C-3, D-4

(D) A-2, B-1, C-4, D-3

Q49. Assertion (A): The direction of Westerlies in the Southern Hemisphere is much more constant and stronger than in the Northern Hemisphere.

Reason (R): The Southern Hemisphere has a vast expanse of ocean with little landmass interference.

[SSC CGL Tier-I 2023]

(A) A is true but R is false.

(B) Both A and R are true and R is the correct explanation of A.

(C) A is false but R is true.

(D) Both A and R are true but R is NOT the correct explanation of A.

Q50. Which of the following statements about the 'Walker Circulation' is NOT correct?

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- (A) It involves trade winds blowing from east to west across the Pacific.
- (B) During an El Niño event, the Walker Circulation strengthens significantly.
- (C) During normal conditions, it brings heavy rainfall to the western Pacific (Indonesia, Australia).
- (D) It is an ocean-based system of air circulation in the equatorial Pacific Ocean.